

DS 289: Numerical Solution of Differential Equations

Assignment 1

Instructor: Konduri Aditya

Due date: **12 Feb 2026**

Total points: 100

Please follow the instructions below in preparing the solutions:

1. Provide solutions in the same order as questions.
2. All the codes should be in C/C++/Fortran. Use Matlab or Python for plotting graphs.
3. The report should be in a PDF format with the necessary steps, plots, explanations and discussions for all the plots.
4. Compile all the solutions, including graphs, into a single PDF file.
5. Create a separate folder for each question that involves a code. Provide the code, makefile, input and output files, which are used to obtain the solution, in the folder.
6. INSTRUCTIONS FOR SUBMISSION PROCESS WILL BE POSTED ON TEAMS.
7. Code submissions will be checked for plagiarism. Copying from external sources, including the internet, is prohibited.
8. You may use LLMs and AI tools to generate code with proper acknowledgment, but using them to write the report is not allowed.

Questions

1. (a) For a function $u(x)$, derive a second-order accurate finite difference approximation to compute d^2u/dx^2 at a point i on a non-uniform grid. The scheme should use a one-sided stencil to the left of i . **(5 points)**
(b) Compute first-order and second-order derivatives of $u(x)$ at point i in a uniform grid, considering a cubic profile of the form $a_0x^3 + a_1x^2 + a_2x + a_3$, over 4 grid points: $(i-2, i-1, i, i+1)$. Also, find the leading term of the truncation error and comment on the order of accuracy for both cases. **(10 points)**
2. Consider the initial value problem over the interval $t = 0$ to 2.0 where $y(0) = 1$. **(25 points)**

$$\frac{dy}{dt} = 2ty - t^3y \quad (1)$$

- (a) Solve this ODE analytically and plot the solution.
- (b) Obtain the numerical solution for the step sizes = 0.25, 0.125, 0.0625, 0.03125, and 0.015625 using:
 - i. Forward Euler method.
 - ii. Backward Euler method.
 - iii. Fourth-order RK method.

Compare the solutions from each method using step size = 0.015625 with the analytical solution in a single figure. Next, plot solutions for all step sizes from each method along with the analytical solution (make a subfigure for each method).

- (c) Compute the maximum and average absolute errors for each step size using the analytical solution and obtain the error graph demonstrating the order of accuracy for each method. Note: Use separate graphs for maximum and average errors to compare all three methods.

3. The equation of motion for a damped pendulum with external forcing is: (20 points)

$$\frac{d^2\theta}{dt^2} + b \left(\frac{d\theta}{dt} \right) + \left(\frac{g}{L} \right) \sin(\theta) = A \cos(\omega t) \quad (2)$$

where, $\theta \rightarrow$ angular displacement (rad), $t \rightarrow$ time (s), $b = 0.5 \text{ s}^{-1}$ (damping coefficient), $g = 10 \text{ m/s}^2$, and $L = 1 \text{ m}$. The initial conditions are $\theta(0) = \pi/4 \text{ rad}$ and $\frac{d\theta}{dt}(0) = 0 \text{ rad/s}$. The forcing parameters are $A = 0.8 \text{ rad/s}^2$ and $\omega = 1/3 \text{ rad/s}$.

- (a) Solve this equation using the RK4 method over the time period $0 \leq t \leq 30\text{s}$.
 - (b) Starting with time step $\Delta t = 0.4 \text{ s}$, progressively reduce the time step by half until you achieve grid independence. Plot angular displacement θ vs. time t and angular velocity $d\theta/dt$ vs. time t , and discuss your observations.
 - (c) Plot angular velocity ($d\theta/dt$) vs. angular displacement (θ) in a figure, which is called a phase-space diagram.
4. Solve the ODE that describes the temperature distribution in a circular rod with internal heat source (S) using finite difference methods (20 points)

$$\frac{d^2T}{dr^2} + \frac{1}{r} \frac{dT}{dr} + S = 0 \quad (3)$$

over the range $0 \leq r \leq 1$, with the boundary conditions $T(r = 1) = 1$ and dT/dr at $(r = 0) = 0$.

- (a) Solve this equation using the second order central difference scheme and find temperature distribution along the radial direction for $S = 100, 500, 1000, 1500$. Use a grid size of 1024 points.
 - (b) Plot the temperature distribution for each case and discuss the results.
 - (c) Compare these plots for different S and find the maximum S such that the peak temperature in the domain does not exceed 200.
5. Consider the following two-dimensional Poisson equation, (20 points)

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = A \sin^2(2\pi x/L) \sin^2(2\pi y/L), \quad (4)$$

where $A = 10^5$, $L = 1$ and $0 \leq x \leq L$, $0 \leq y \leq L$. The boundary conditions are illustrated in Fig. 1. Use central difference method to approximate the derivatives and iterative methods, such as Jacobi and Gauss-Seidel, to solve the linear system of equations. Consider the resolution $\Delta x = \Delta y = 0.01$ and relative tolerance 10^{-6} to obtain the numerical solution.

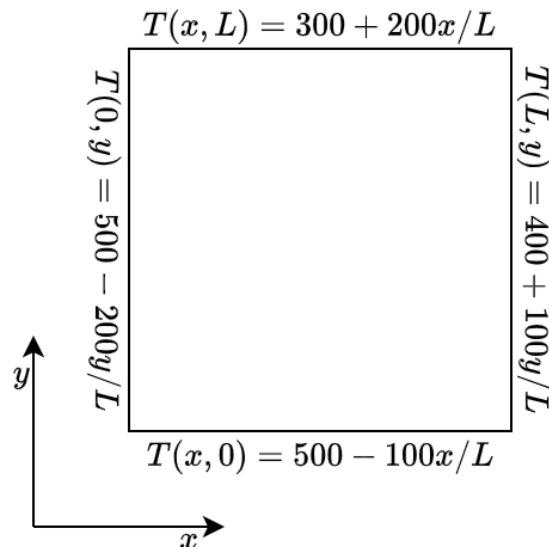


Figure 1: Boundary conditions for 2D Poisson equation problem.

- (a) Plot the distribution of the converged solution $T(x, y)$, obtained from both Jacobi and Gauss-Seidel methods. Next, plot the variation of T along the line corresponding to $x = 0.25$, $0 \leq y \leq L$. Then, plot the relative error vs iteration for both Jacobi and Gauss-Seidel methods, and discuss your observations.
- (b) Solve the same problem using the successive over-relaxation (SOR) method. Consider over-relaxation parameter ω of values 1, 1.2, 1.4, 1.6, 1.8, and 2. Run the iterative algorithm for 500 iterations for all ω values and plot (i) the error vs iteration for all ω , and (ii) the error at the end of 500 iterations vs ω . Comment on the trends based on your observations.

NOTE: For consistency, use the same initial guess solution across all the methods while performing the analyses.